

Arc Length and Improper Integrals



Arc length

- 1 Find the arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.
 - 2 Set up (do not evaluate) the arc length integral for $y = \sin x$ on $[0, \pi]$.
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Improper integrals

- 3 Evaluate $\int_1^{\infty} \frac{1}{x^2} dx$, or state that it diverges.
- 4 Evaluate $\int_1^{\infty} \frac{1}{x} dx$, or state that it diverges.
- 5 Evaluate $\int_0^1 \frac{1}{\sqrt{x}} dx$ (improper at the lower limit).